

BreathCoach

On-device breath detection for singing

THE PROBLEM

Breath support drives singing — where you breathe, how long your phrases run. Singers get no objective feedback on it.

THE GAP

No deep-learning breath detector built for singing.

No open breath-labeled singing dataset to train one.

Data

USED

GTSinger

singing corpus with breath tokens → training

VocalSet-Breath

Open breath labels on VocalSet (CC BY 4.0).

250 labeled

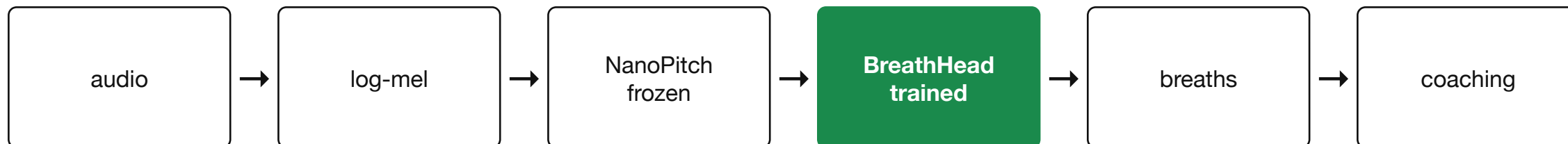
Baselines

Held-out singer · event-F1 @ ± 200 ms

Method	F1	Recall
Ruinskiy & Lavner 2007 (DSP, no learning)	0.01	0.4%
BreathCoach (ours)	0.67	97%

Ruinskiy 2007 — matches audio to a hand-built breath template (no learning). **Respiro** — deep model, but trained on speech.

Innovation



Only the 15K-param head is trained, on a frozen 333K pitch model. Runs on-device (CoreML, 468 KB).

LIMITATIONS

- Batch, not real-time (yet)
- Coaching = phrase length, not a vocal-support score



Demo